THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 10: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (5x + 2)$ cm and $QB = (3x + 12)$ cm. The length of QA is: 1

(a) 29 cm **(b)** 37 cm **(c)** 5 cm **(d)** 27 cm

2. If a tangent PQ at a point Q of a circle with centre O makes $\angle OPQ = 30^\circ$, then $\angle POQ$ is: 1

(a) 60° **(b)** 30° **(c)** 120° **(d)** 90°

3. A point P is 39 cm from the centre O of a circle of radius 15 cm. The length of the tangent drawn from P to the circle is: 1

(a) 36 cm **(b)** 24 cm **(c)** 54 cm **(d)** 1296 cm

4. A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 8$ cm, $BC = 13$ cm and $CD = 10$ cm. The length of AD is: 1

(a) 15 cm **(b)** 11 cm **(c)** 31 cm **(d)** 5 cm

5. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle APB = 90^\circ$. The $\angle OAB$ is: 1

(a) 90° **(b)** 20° **(c)** 45° **(d)** 30°

6. A tangent to a circle of radius 30 cm has length 40 cm from the external point P to the point of contact. The distance of P from the centre of the circle is: 1

(a) 100 cm **(b)** 10 cm **(c)** 70 cm **(d)** 50 cm

7. In $\triangle ABC$, right-angled at B, $AB = 8$ cm and $BC = 15$ cm. A circle is inscribed in the triangle. The radius of the inscribed circle is: 1

(a) 6 cm **(b)** 4 cm **(c)** 3 cm **(d)** 8 cm

8. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (2x + 7)$ cm and $QB = (x + 12)$ cm. The length of QA is: 1

(a) 17 cm **(b)** 5 cm **(c)** 22 cm **(d)** 19 cm

9. A quadrilateral $ABCD$ is drawn to circumscribe a circle. Which of the following is true? 1

(a) $AB + BC = AD$ **(b)** $AB + AD = BC + CD + 1$ **(c)** $AB + CD = AD + BC$ **(d)** $AB = CD$

10. A point P is 40 cm from the centre O of a circle of radius 24 cm. The length of the tangent drawn from P to the circle is: 1

(a) 32 cm **(b)** 16 cm **(c)** 64 cm **(d)** 1024 cm

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 10$ cm, $BC = 6$ cm and $CD = 9$ cm. Find the length of AD. **2**

12. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (7x + 2)$ cm and $QB = (4x + 14)$ cm. Find the length of QA. **2**

13. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle APB = 120^\circ$. Find the $\angle OAB$. **2**

14. A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 16$ cm, $BC = 12$ cm and $CD = 7$ cm. Find the length of AD. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The incircle of $\triangle ABC$ touches BC, CA and AB at D, E and F. $AB = 15$ cm, $BC = 13$ cm and $CA = 14$ cm. Find the lengths AF, BD and CE (in this order). **3**

16. In $\triangle ABC$, right-angled at B, $AB = 10$ cm and $BC = 24$ cm. A circle is inscribed in the triangle. Find the radius of the inscribed circle. **3**

17. Two concentric circles have radii 25 cm and 7 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. **3**

OR

The incircle of $\triangle ABC$ touches BC, CA and AB at D, E and F. $AB = 11$ cm, $BC = 8$ cm and $CA = 9$ cm. Find the lengths AF, BD and CE (in this order). **3**

18. In $\triangle ABC$, right-angled at B, $AB = 27$ cm and $BC = 36$ cm. A circle is inscribed in the triangle. Find the radius of the inscribed circle. **3**

OR

Two concentric circles have radii 26 cm and 10 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that the tangents drawn at the ends of a diameter of a circle are parallel. **5**

OR

In $\triangle ABC$, $\angle B = 90^\circ$, $AB = 12$ cm and $BC = 16$ cm. A circle touching all three sides touches AB at F and BC at D. Find the radius of the circle and the lengths AF and CD (in this order). **5**

20. Prove that if a circle touches the sides BC, CA, AB of $\triangle ABC$ at D, E, F, then $AF + BD + CE = \frac{1}{2}(AB + BC + CA)$. **5**

OR

In $\triangle ABC$, $\angle B = 90^\circ$, $AB = 7$ cm and $BC = 24$ cm. A circle touching all three sides touches AB at F and BC at D. Find the radius of the circle and the lengths AF and CD (in this order). **5**

END OF QUESTION PAPER — Total: 40 Marks